

Precision Convergence Webinar Series with Dr. Karl Friston

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so uh for those who are joining uh i am laura dubrey the co-chair of this webinar uh we have two more minutes before we formally start we will be officially starting at 11.03 [Music] ah there's lauren and alan that's not laying here yet i forgot about elaine yes he's there i see him he's online okay good yeah we're up to 80 participants dr dagger it's uh good to see you seen you in a while yes same here we all meet in cyber space those days okay thirties uh we will be starting uh just now i think uh it's close to 1103 good morning everyone welcome to this webinar that is part of the precision convergence capacity building webinar series this series is co-hosted by miguel center for the convergence of health and economics and the pittsburgh super computing center the series was started about a year ago in the following up the hpc high performing um computer 19 consortium that had been engaging psc and other research computing institutions around the country with ibm and many of the other data business uh for accelerating our ability to bring solution to covet 19. this consortium was created for seeing to what extent we can bridge the omics and all the digital capacity to accelerate solution to covet 19 profiling uh characterizing of the virus vaccine and so on and so forth and we said uh what if we were to take as a silver lining of the kobe 19 continue building of this capacity of ai and digital technology and others to see to what extent we can bridge the various science uh that are needed if we are to bring uh in a timely manner a solution to real world problem whether it is pandemics or whether it is climate change and so on a lot of it is tied to the fact that in the last 250 years we have changed a lot the context within which the human machine has evolved the first industrial revolution was uh it is 250 years ago it's essentially a few seconds in the journey of mankind so this series is really to bring the scientists the action leader that are operating on multiple scale in saying can we go beyond than what we are each doing so far so before i introduce uh more what is it we'll do together during this uh next two hours and the keynote um i uh would like to ask my co-chair to introduce himself and then i will come back

for introducing uh the keynote go ahead sean hi my name's uh sean brown i'm the director of the pittsburgh super computing center a joint computational research center between carnegie mellon university and the university of pittsburgh and i'm also the vice chancellor for research computing at the university of pittsburgh i'm very excited about our talk today and our panel follow-up panel discussion so i will take less i will not take any more time so thank you all for joining us thank you sean uh very briefly in a nutshell the way we spend the next two hours um i have been for those of you who were the on the series before we always start by saying it's not a webinar it's a capacity building webinar where we convene a keynote of significance uh in advancing science uh in one of the three domain the human machine the man-made system or the computer and we build a panel where everyone is carefully selected to not only be aware and complementary to the keynote but each of the panelists as you can see are also bringing those uh the their own perspective in advancing this precision uh convergence program uh so we have the first uh 30-40 minutes of the keynote presentation and then we have a first round of panel where each of the paralysis bringing a three to five minutes commentary from their perspective in response to the keynote presentation but also from their own uh their own uh knowledge and so on um the keynote has about two or three minutes for each of them to answer in the first round uh and then uh in the second part then if there are questions uh from the audience sean will be managing them from uh the chat room uh to address them after the first round of the panelists we have a second round where everyone including the kingdom uh is invited uh to kind of do the one minute commentary of where can we go as a world community if we are to really reach a scale uh into the transformation that we are all wanting to achieve okay uh so uh this is um uh this is kind of the logistic aspects uh i would like before introducing uh dr carl christian to position a bit his presentation within the continuum of the webinar we have been having over the last year on the psychology south side we have been having as kino roy bom mister um daniel uh schachter uh the last one on which carl was on the panel was by constantine sediquides all of them uh really one can arguably say uh the most influential living scientists in the study of psychology and the mind and the and and so on um and uh we have been and in the last webinar where carl was presented was on the panel the discussion was to see um to what extent the cell to what extent the cell the self and the end the psychology could be placed on the same platform of the whole biology when we look at a human being being evolving not only for survival but also for achievement and this is within that context that have been uh discussing and that's why the work of the the work that will be presented today on the self evidencing for individual and collective agency to what

extent can this substantive knowledge of human uh biology psychology and real world can be brought together in some degree of formalism that help bridging all of the above so that is why without further introduction i'm extremely pleased to um to welcome carl freestone and as we call as shawna said uh we are doing introductions very short because there is um there is a limited time with the richness of the intellectual and human capital we have together in the next two hours so calfstone is a theoretical neuroscientist and authority on brain imaging i won't enumerate the various statistical methods that he has been developing uh his most its main contribution uh to theoretical neurobiology is a free energy principle for action and perception which is the focus uh of uh his style today so carl if you excuse me i will not give a read the whole uh the whole um description of of your achievement and i will go with you go to you directly for your well thank you very much it's a great pleasure to be back here and it's now my turn to do a keynote of significance i've got to do that with two constraints first of all get everything that's important on the table within half an hour and also use not my favorite words but the favorite words of my panelists so we've all met um last week just to sort of cover the key themes that we want to bring out in discussion so my job is to try and situate those um within this keynote and this keynote's um going to address the first principle account of sentient behavior so a story about how i behave and how we behave that can be elaborated uh from the first principles of self-organization and the um approach i'm going to take is from the point of view of self-evidencing so i'm going to introduce the notion of self-evidencing as simply as possible motivate why this might be an interesting approach in relation to current perspectives on artificial intelligence machine learning um and then try to ground it in the neurobiology as was um you know my brief or our brief in the introduction um i'm going to take a special focus on the mechanics of belief propagation and in particular how we organize our behavior to solicit the right kind of data that supplies evidence for our own existence and sell that notion that imperative as an important drive for um active engagement with the world and then finally uh talking to the collective aspect and the importance of collective agency and shared narratives so self-evidencing what do i mean by that and i simply mean that everything that we think or do can be understood or described in terms of trying to maximize moral evidence so what's moral evidence it's simply the probability of some observable outcomes given my model of how those outcomes were generated and on this view we can associate belief updating with perception where i'm denoting beliefs here as with cue of some hidden states or causes latent states of the world that are generating these outcomes and in a similar vein i also i'm going to assume that we have beliefs about what we're going to do about our

actions and prior to selecting a particular action we have to form beliefs about what is the most likely action you that i'm going to commit to and the argument here is that both of these these kinds of leap updating are driven by an imperative to maximize model evidence so that has a degree of construct validity in relation to established normative accounts of behavior um i've listed a few here if i read this log evidence as the value of an outcome in the sense that if i prefer these kinds of outcomes or these outcomes are characteristic of me then i should behave to secure those kinds of outcomes as valuable outcomes and from that we can spin off uh if i was in psychology reinforcement learning i was an engineer we could cast this as optimal control theory and indeed if i was an economist it would just simply be one simple specification expected utility theory if on the other hand i was an information theorist i would read this quantity or the negative of this function this negative log probability of an event or an outcome uh conditioned upon a model as a self-information sometimes called surprisal or simply surprise and it is this quantity that is uh i'm going to associate with um attractable measure of the value or the negative value or the surprise namely a free energy and this quantity suggests that everything we do is in the service of minimizing surprise we can read that as the principle of maximum mutual information on the max principle uh principles of minimum redundancy or maximum efficiency that can be attributed to horus barlow and indeed the free energy principle which is a formalism i'm going to use this is nice because the expected self-information expected surprise before an event has happened is entropy so this also says that we perceive and act in a way to minimize on average entropy and of course that's the holy grail things like self-organization and physics energetics and if i was a physiologist it would simply be a statement of homeostasis that i am in the game of keeping my outcomes within some viable bounds minimizing their dispersion or the dissipation i.e their entropy and if we are a statistician i would treat this quantity as the model evidence and from that we can spin off things like the bayesian brain hypothesis evidence accumulation and various implementations of that such as predictive coding so i'm going to unpack this quantity this surprise in terms of value in terms of a free energy functional and try and give you a number of different perspectives that um on this quantity so the first perspective is that we can regard this statement as a first principle statement about the imperatives for self-organization that inherits directly from james's maximum entry principle so in maximizing um this free energy functional here we are effectively increasing the dispersion of our beliefs we're keeping our options open to the greatest extent possible under some constraints and i hope we're going to talk about constraints later the particular constraints here are afforded by something called a

generative model that is usually decomposed into a likelihood and a prior the likelihood of some outcomes given their causes latent states in the world and some prior beliefs or preferences about the states that I am likely to be exposed to in terms of generating my observable outcomes there's another way of writing down this quantity which provides a complementary perspective and that's the kind of perspective you get if you're a statistician so a little rearranging the terms in that free energy functional and to express it as a mixture of accuracy and complexity so the accuracy here is simply the expected log likelihood outcomes given my beliefs about how those outcomes were formed whilst the complexity is a very interesting term in this setting it's just the difference between my beliefs after and before seeing some observations or evidence for beliefs about particular states of the world so it literally scores the degree I change my mind when I see an outcome or I observe something it scores the cost of belief updating and the story that I'm going to tell for the remainder of this talk is that if I have to choose what to do that will get some more information get some more observations then I can cast behavior of a sentient agent in terms of choosing those actions that maximize the accuracy and minimize the complexity I expect following that action and it turns out that the expected accuracy complies exactly with the principles of optimal foraging in design while the optimum the expected complexity turns out to be risk and is exactly the quantity that underwrites uh optimal decision theory optimal bayesian decision theory so what I'm saying is that putting these two things together gives us self-evidencing in the spirit of um selecting those behaviors or actions that maximize this expected free energy and implicitly um increase the expected evidence so just to make this insight more heuristic and to compare and contrast with alternative approaches imagine you're an owl and you're hungry and if I was there with you I would be asking somebody on the front row what are you going to do and invariably they give me the correct answer well I'm going to look for my food I'm going to search for my prey and that simple answer tells us something quite fundamental about the way that you would write down a calculus for sentient behavior um and the two kinds of calculus I'm going to compare and contrast and then bring together um later on um rests upon the notion of a value function of states of the world I could say okay if I had some function that scored the goodness the value of a state that ensued following at the next time point following a chosen action u I could then just choose for every given state of the world the best or the the value function maximizing action to give myself a state action policy is this kind of calculus apt for describing behavior well the answer I'm searching for my food tells you no it can't be because searching is an act that reduces your uncertainty about the location of your brain but

uncertainty is not a function of states of the world it's a function of the beliefs about states of the world and so we have an alternative way of writing down good behavior and that is something that maximizes a functional function of a function in this instance that function is beliefs about the states the world that would ensue if i committed to a particular action it's a very different kind of functional form the other aspect of searching for food tells you something also about the nature of the best way to describe or the best policies and that's it matters whether i search my food and eat it or try to eat it and then search for my food that tells one immediately that what we're um chasing here is a way of describing a sequence of actions or actions over time and this functional then becomes technically a path integral um a sum of goodnesses over time and that gives you a very different functional form for the optional policy it's now a sequential policy that maximizes this uh path integral of an energy functional of the senior moment it's a free energy functional the reason i'm using those words is just to refer to the two kinds of principles that we could appeal to in underwriting these different approaches to good behavior so this state action policy is predicated on belmont's optimality principle we'll find that in optimal control theory dynamic programming deep reinforcement learning basic decision theory and so on this on the other hand has a slightly deeper and simpler heritage it's it's basically a variational principle of station reaction um the free energy principle being one instance of that um we will see later that this is more apt to describe things like artificial curiosity or in robotics intrinsic motivation and is more concerned with data gathering optimal bayesian design than not based in decision theory and i repeat is concerned with optimizing sequences of policies under uncertainty using things like partially observed markov decision processes so principles of least action um this is the and i apologize for the the equations here there's a beautiful symmetry um between the functional forms of this way of writing down evidence in terms of complexity or accuracy things the equivalent things that you are optimizing if you are selecting those behaviors that you think will maximize the expected free energy just writing down again an expression for the free energy here in terms of accuracy and complexity and rearranging it in a particular way here that people in machine learning will recognize so just switching these terms around we can also express this as the log of the evidence the probability of some outcomes given a model plus something a kr divergence something that is never less than zero because this thing is never less than zero this quantity this free energy now becomes an evidence bound uh known by the acronym and elbow and evidence lower bound machine learning so given these two ways of carving up this objective function um what would they look like in expectation well

what happens is that the expected complexity becomes risk the expected inaccuracy becomes ambiguity the expected log evidence on negative log evidence becomes expected cost and the expected divergence becomes expected information gain and this is beautiful but it's also quite unremarkable because uh most people in one field or another already know about this so for example if we ignore um the expected value or the negative expected cost then we're just left with this expected divergence what is this well it's simply the difference between my beliefs about states of the world in the future following a particular policy if i were able to observe the consequences of that behavior relative to if i were not so it scores the information gain or the reduction of uncertainty about hidden states or latent states of the world afforded by this particular policy so it's the epistemic affordance of this policy and the visual search literature this will be known as expected bayesian surprise more simply it's just the um the mutual information or the expected mutual information conditioned on a on a policy i'm going to successively remove sources of uncertainty from the table and just see what emerges and what's going to happen is we'll get back to the development optimality principle and expected utility theory but on route um let's assume that we live in a universe there's no ambiguity states or outcomes and outcomes and states i can see everything directly and this means that we can ignore the ambiguity term and supplant the s's for o's and vice versa we're just left with this which is another kl divergence and it just scores the difference between what i anticipate will happen if i pursue this policy and what a priori i prefer to happen given i am me this is the expected the value or the negative um the negative uh cost um expresses the probability and so this minimizing this difference this divergence between what i anticipate and what i prefer is known in engineering as kl control and economics it's sensitive control and it's simply a description of behavior in which you choose those things to do which minimize the divergence of the outcomes that you expect under a generative model a mixture of likelihood and prize um namely your pride preferences and then finally if we remove the kind of uncertainty that is reducible by my epistemic foraging or responding to epistemic affordances or the intrinsic motivation for finding out what would happen if i did that then we could ignore the expected information again we're just left now with the expected cost and that clearly um can be read now as expected utility theory if you're in economics so in summary what we're saying is that there's a simple way of writing down self-evidencing that speaks to engagement with the future and the way that we choose what to do next and we can decompose maximization of expected free energy uh which we refer to as active influence in terms of two dual aspect bayesian or dual aspect basing optimality one in terms of maximizing expected value or

minimizing expected cost uh in accord with um bayesian decision theory based on making base optimal decisions and at the same time we're maximizing expected information gain and this inherits from the work of people like dennis lindley um on optimal basic design basically designing your experiments in a way that you solicit those data that resolve the greatest amount of uncertainty about your hypotheses or models of how data are generated um so to close i'm just going to race through things you can do with this first principle like account or method one the first thing that one can do obviously is to simulate and uh um in silico reproduces agents and artifacts and see how they behave but to do that you have to commit to a generative model these are the constraints that when supplied to james's maximum entropy principle gives you the free energy principle so one key thing is that if you can write down a generative model you can always express it in terms of a probabilistic graphical model what this slide is meant to convey is that then there always exists something called a factor graph so why is this important well the factor graph is just if you like a dual or an adjunct way of expressing your generative model in a way that reveals immediately the nature of the messages that need to be passed amongst the different belief structures that are over the random variables the hidden causes um that generally are organized according to this markov decision process so that we have um outcomes that are generated by states of some universe those states generate outcomes through a likelihood mapping that's a likely part of my generative model but the states evolve over time through transition matrices that constitute my prior beliefs about the dynamics and the contingencies and the laws um that denote the causal structure in my universe and it is these transitions that i'm in charge of uh in terms of the policies that i select and i select my policies to maximize the expected free energy here so that would be a generic graphical model description of an active engagement uh with the world and this would be the equivalent factor graph that tells me the message passing that i would need to do to do the brief updating to realize and integrate this system to simulate um self-evidencing or active inference and there are all sorts of great games you can play here in terms of once you know the message passing the uh the belief the the form the functional forms and schemes of belief propagation that's in some way must be under the hood for any given generative model you want to commit to and you can start to think about how this would um what this would look like in various systems such as the brain i won't go into this but there's a again a very beautiful and very simple mapping between the time scales of different updates during say belief updating a perceptual inference slower updating of the parameters of my generative model that we can associate with learning and action selection that ensues from this notion of planning as

inference in terms of updating beliefs about what i'm going to do next and we can map this uh these functions onto different brain areas and functional brain architectures and make predictions about the nature of those architectures which i'm sure we'll be talking about later we can simulate behavior i'll whip through uh one example just to sort of demystify and just pick out the most important um aspect of this of this formulation which is basically this um this balance um this dual aspect optimal bayesian design versus optional base in decision theory um and show that in fact that balance is systematically deterministically specified by how much i know about my environment effectively it dissolves the exploration exploitation uh dilemma um and to illustrate that there's a little game that we've created here we're simulating this mouse and the game there for this mouse is it has to in just two moves secure its reward that could be on the left or the right of the teammates and so it could either take a chance and find out where the reward is it has to stay in one of the upper arms um if it chooses one so you know half the time it may be right and spend two bits moves with its reward the other half of the time it will be wrong and has to stay with the non-rewarded in the non-rewarded location for its two moons but it has another choice and this is the important choice from the point of view of this little demo we've also equipped this universe with a deontic queue an instructional cue that if it uses one of its moves to actually go and see the color of the queue down here this q tells it where the reward is so now it can choose to use the first move to explore to resolve uncertainty about where the reward is and then the second move to go with a degree of confidence to secure its reward um now clearly from an expected utility point of view these are equally um viable options but from the point of view of self-evidencing with this epistemic uh affordance in play it's going to be much more attractive for this little mouse to go and resolve uncertainty by going straight down to the uh informative view when you simulate this mouse under a particular giant model using those update equations i showed you a couple of slides ago that's exactly what happens um so we start off uh with the little mouse going down finding where the reward is and then sitting on its reward for the final move um but as time progresses these details are unimportant the only thing we're trying to convey here is as time goes on what we've done here is leave the reward always on the left-hand side so that this mouse now accumulates the belief that reward is always in the same place and as it gets more and more confident it accumulates more evidence that the reward is always in the same place the epistemic affordance or attractiveness of the instructional cue starts to weigh and the epistemic value the epistemic affordance will at some point become smaller than the expected utility or the expected cost um of continuing to forage epistemically and at which

point the behavior reverts to explorative exposure behavior where it will go straight to where it now knows the reward is and stay there for two minutes and we can see that occurring after about 20 trials in this particular example that switch in behavior affords the opportunity now to do all sorts of interesting in silico experiments and make predictions about either neuronal responses to the chosen versus unchosen option um or one can look at the encoding of behaviors and locations in terms of place cell activity and indeed we can simulate this negativity paradigms or ball paradigms by just comparing trials where at the beginning there was lots of uncertainty so everything is novel everything is an audible with trials with exactly the same behaviors and sensory stimuli or sensory outcomes but now after the mouse has become familiar no more was bored with its environment so everything now becomes a standard and again we can uh reproduce with some marker remarkable fidelity the timing and the from the forms of simulated electrophysiological responses so i'm giving these examples very very quickly just to show that they both grounded in and to a certain extent inspired inherit from the neurobiological implementation of this this sort of belief updating scheme so i'll finish now with a um with an important nod but it'll be no more than a nod so what does this kind of formulation bring to the table when we're thinking about lots of agents um acting together uh oh um succumbing to the same kinds of imperative and one really interesting sort of thing that emerges technically from this is that we move from a world in which we're trying to think and explain the behavior of a creature or a particle that's making inferences about it the states of the world out there the environment through um observable things and uh over which it has control through active things and we refer to the these two kinds of variables as a markov blanket we move from a structure where there's a markov blanket separating the internal states that do the belief updating um from the outside world of the supplying sensory evidence and is subject to our actions upon which we act if we now think about two of these agents putting together then in fact the external states disappear because you become my external states and i become your external states so there are no more external states there's just lots of internal states my brain states your brain states mutually trying to infer each other and to cut a long story short um the um the the natural consequence of this in terms of both of us trying to minimize our is that we will learn to make ourselves more predictable and we will query each other in a way that we will resolve uncertainty um about you are i will resolve uncertainty about you um but i can only do that if we have the same shared generative model um i'm going to close with an example where we use the intrinsic motivation the epistemic affordance to compel two agents one of which can see a scene and the other one can't but they

can talk to each other and because there's this epistemic affordance or incentive to resolve uncertainty one agent knows because it shares the same language with the other agent that she knows what questions to ask to resolve uncertainty about the scene that she cannot see but the other agent can and it's quite simple just to write down the generative models and the equivalent graphical uh factor graphs um for a very simple world where two people are basically uh playing well they're playing 20 questions and one person asks is is is something true or is something like that and the person say yes or no and they will slowly share their beliefs so in this example um the um the blue um woman can see two shapes two blocks there's a red square below and there's a green square above this lady the teal lady cannot but she can ask questions so you can ask this lady is there a triangle below the stage says no there isn't um and then this lady knows that there is a square but does not know the color and through this process of making the right moves um engaging in policies that resolve the most uncertainty to by maximizing the she asked the right question about the color of the square below and finds out that it's not green so it has to be red and then she can drill down on the um the nature of the upper object until after about four questions she knows exactly what the other person is looking at in fact they can switch roles and uh the other person can check that the first person has got the right idea and so the reason that i i bring that to the table is twofold one to emphasize um the importance of this epistemic foraging in the context of interpersonal interactions and um and the mechanics of exchanging beliefs and the other thing is what underwrites this which is the shared generative model that is necessary to actually communicate things that are very necessary if you want to assimilate belief updating in a distributed way that has to go across a markov blanket in some ensemble or collective um now what's on the other side of the markov bank it doesn't always have to be the same it could actually be a you know a a an eco niche um but still there's going to be exchange of information and querying and when lifted to the cultural level then we could actually talk about these uh this formulation as a one way of understanding cultural niche construction and then finally if we now apply these principles um over different time scales we can also now think about sort of optimizing model evidence through moral selection as a mathematical image of basin um of natural selection and evolution um so with that i will close just for the final world to to einstein everything should be as simple as possible but not simpler with a not this important notion of complexity um that is an important part of model evidence so with that uh i just want to thank those people whose ideas i've been talking about of course thank you for your attention thank you very much indeed thank you very much carl for this brilliant

uh presentation that get as simple as this can be in presenting that complexity and i really also appreciate that you squeeze our webinar series into your busy schedule after uh the one we have with uh uh with uh constantine i'd like to ask uh a twin question uh before we get to the panel uh uh and that question is trying to my i could say uh let's say sophisticated lay woman or even just lay woman on understanding that i think i'm getting to get to of your mark uh marco that is kind of a core aspect which i believe is um kind of the statistic statistical space that that combine or differentiate what is in an out system if i understand correctly my two questions are the following as you most likely don't know i was throwing 30 years ago at colonel at the time that behavioral economics was starting with the um with uh zversky kahneman and uh dick taylor that was very uh descriptive uh uh vision of the non-rational side of human behavior uh in moving toward precision convergence i progressively get much more around along the work of herb simon that is defining essentially bounded rationality as being resource rational within the context of your system so my first question is to i'm intrigued to hear you connect your work with herb simon's view of bundled rationality and his approach and within that context also on the panel that we were having on on when consulting was presenting is homeostatic view of self-promotion and there was this debate between this model whether the psychological self-promotion promotion and the the biological protection for immunity and no one was trying to seize their cause a common causal mechanism so i would be intrigued in my question my second question if you could briefly comment on advancing the discussion we were having between the psychological the mind and the self and then the whole biological business within the context and the challenge is answering this in three minutes go i'll spend the first two on the first questions excellent questions service yeah simon's unbounded rationality i love that because of course um there's a nice semantic clash with an evidence bound so if you go to machine learning and ask how does a variation auto encoder work it optimizes its weights with respect to this elbow so it's your it's all in the bound so bounded rationality approximate bayesian inference defined operationally as optimizing a variation free energy literally is is bounded uh optimality and the the other thing that's nice about it is this notion of constraints uh and i'm using that in the in the sort of json sense but of course an important constraint is the uncertainty under which you're operating so you know that's why i was so keen to emphasize that you can tease apart the overall objective function um into two bits but remember one bit is contextualized by the others so the base optimal rational decision making the under the um the the expected utilitarian is always contextualized by a degree of uncertainty so there's a value of information that provides if you like a a space

or a constraint or a context for and that and you're i'm i'm reading that in the spirit of bounded rationality where this is a computational resource or an information resource and with respect to the homeostasis yeah i mean i love constantine's stuff because you know it's just lifting this basic imperative for a generalized homeostasis a generalized synchrony with your environment from the physiology to the self-modeling to the things that underwrite our behavior and speak to this really important notion that if it is the case that we behave in order to secure or realize our the kinds of outcomes that we prefer then just to be base optimal in this generalized sense is basically to have that sampling bias and just to to be to be intrinsically biased and if you weren't you wouldn't exist you know if you bring about the things that you a priori expect and then that is a sufficient explanation for why you conserve yourself over time and of course that's just a statement of homeostasis thank you so much and so i would like to ask the panelists to all uh open their video shifting to sean my co-chair who can ask his privileged question also and start uh the panel go ahead sean no i'm not i'm not going to take that privilege because i'd like to hear what our panelists i'm very interested what our panelists have to say so we'll go ahead and get it kicked off with our first round of commentary i'd like to introduce mark bailey who is the vice president for re uh vice president for research at cfar and a professor in the department of computer the departments of computer science statistics and actual science electrical and computer engineering mathematics epidemiology and biostatistics and biology at western university so welcome mark and looking forward to your commentary thanks sean and uh thank you thank you carl for an absolutely beautiful presentation so you asked in your opening slide is there some information theoretic imperative that renders inference and decision making two sides of the same coin and then you've given us this elegant and and complex model of how precisely that that might be implemented and uh i'm a simple person so i needed a simpler questions i i i said i'm going to step back and ask how might such a mechanism arise why might we end up in this place and i suspect most of us here in the audience have had the experience of you're using your laptop and it's doing a computation or a program's crashing and suddenly it gets very hot and you know what why is that why is your laptop getting hot when it's working harder it's not making toast it's just moving bits around and so there's a fundamental disconnect because we still tend to think of the physical world in terms of mass and energy but you know since the time of john archibald wheeler we know it's actually mass energy and information so information is a fundamental uh fundamental core object in our physical universe and so wheeler had a tagline it from bit you know from a bit of information we can create something physical and in fact that bit of

information is something physical and then in the 1960s rolf landauer at ibm watson labs conjectured that erasing information is thermodynamically dissipative so if you take a bit of information and you erase it you clear that bit that will release this is controversial at the time there's literature both supporting it and not sporting it but there's been a spate of experimental results recently over say the past two decades that have really strongly confirmed that yes erasing of information is dissipative so information processing and thermodynamics are linked so if you if you take this framework and what we know in theoretical computer science i'm going to say something is not quite true so forgive me the harder the computation the more deletion is involved in that computation and the more energy you need we could spend hours talking about what i mean by hard what i mean by needing deletion but just trust me on this one for now so the question i come back to is would evolution select against unbounded computational complexity which is to say unbounded energy requirements and i think that that that's reasonable to say it would be parsimonious for for evolution to not prefer systems that require unbounded amounts of energy and so this question has been asked in the literature uh martin novak who is a famous evolutionary game theorist has a nice publication in pnas about 10 years ago on the computational complexity of evolutionary dynamics and more recently and on topic for for today uh kaznichiv at oxford has a beautiful paper in genetics with title computational complexity as the ultimate constraint on evolution love it definitely worth reading so back to the brain and to the presentation we the keynote we've just had the brain is touring complete of course any computable function i can compute with my brain give me a pen and a and this big enough stack of paper i can simulate a turing machine but the real question is during my my everyday life whether i'm a mouse in a maze or or me giving a response to a keynote here do i spend most of my time in that regime of super linear recruits recursively enumerable functions so am i really doing hard computations most of the time and and the answer is i'm not right we've all had that experience of driving on autopilot and there there you know there's a lot of systems that are working with very little energy there's there's not a lot of uh of of hard computation going on and so what if the strongest constraint as kennedy chiefs suggests on the evolution of brains is computational complexity where does that get us well to push it i'm going to take an extreme absurd position what if what psychologists call attention what if carl's sentient calculus i love that term by the way what if consciousness what we perceive as consciousness are actually primarily mechanisms for controlling complexity there are mechanisms for controlling computational complexity and everything else all of the richness i experience in my conscious life is a happy side effect of

that imperative that is selected for uh through evolution and you know i think those are actually pretty good mechanisms attention is a great mechanism for for controlling computational complexity of course that you know that was an absurd position to to propose but i i did it to invite conversation in reality there's going to need to be a balance between this need to keep computational complexity low while also providing a robust behavioral repertoire because you do exist in an environment and you must behave in that environment so computational complexity is not the only only item on which evolution is performing selection but i think it's an important one i think the role of thermodynamic complexity of neural computation has been understudied and i i can't prove this but i'm going to think about it harder i think if you pull on this thread you end up with a model that looks very similar to to what carl has just presented here thanks john thanks yeah carl do you have a response just maybe a minute uh response i believe i'd like to respond for about half an hour so many great points there okay in one minute um right first of all selection your natural selection the evolutionary pressure attention selection all one theme i agree entirely it's complex i mean it's only half the story because you've got the accuracy but under under constraints that the accuracy about the same it's all about the complexity so you know you can see that in statistics through bayesian model selection um that you're trying to get the simplest architectures that will generalize so you don't want to over parameterize so even the pragmatic agenda of evolution machine learning is driven by getting the simplest explanation explanation possible so you know there are all sorts of ways in which one could celebrate that you know i'd like to talk about commonwealth complexity and uh universal computation janinsky equality is you know so many right things have been said there but i've only got one minute so i'll shut up thank you i mean to stifle you i just we have a lot of other perspectives to get to so i'll turn back over laura yes and i'm very pleased to uh introduce conrad conrad i apologize in advance for my awful french accent according i discovered conrad's uh work uh in a paper on foraging with the idea of looking at dopamine mechanism that are finding where to go and then where to have to have your your your choice uh so that extra i'm extremely pleased to have him uh with us with his lifetime work on sensory matter and various aspects of normative model in beijing statistic and other he is a german neuroscientist and professor at the university of pennsylvania so i will keep it here conrad to leave you more time go ahead three to five minutes i know you have a lot to say go ahead you bet so so let's zoom out just a little bit now like there's a broader endeavor in our field which is called normative modeling it is it is used in important parts of neuroscience it's also the basis of machine learning arguably what we do

in that field is we say let's try and derive the optimal solution to an existing problem it usually works like that where we look at the world at the problem maybe that the brain is solving or at the problem that we want to solve as machine learners and we write down what our goals are it's usually like a cost function this is what we want to optimize not like get it to be precise most of the time whatever your definition is then we will usually have a set of constraints those things that we cannot deviate from so mark had a great example compute is one of them there's many of many others of them and then there's prior knowledge that we have about the world aspects of the world i think carl carl related to pomdp's earlier and then what we do is we take this problem statement and we kind of derive the optimal solution and if we're a neuroscientist we compare behavior maybe to that or we look for representations in the brain if we're machine learning people we basically just see how good we are at solving the technical problem that we want to solve so in that sense i think we are all sitting in the same boat now where i think there's a rather large disconnect between the way the keynote was thinking about it and the way i think about that field is for me most of what happens is in the details of that interface so when i say goal it's not that simple get it right it's more like on which data set will we get it i want to get it right under which circumstances what will be the future task that we will have to solve with and same thing with constraints it's not just like you only have x computations it's complex all but the world not like it's not just a palm dp world it's a world with structure and causality and like all the different pieces so from my perspective all the interesting things kind of live within these components that we put into that so for example if we want to talk about about curiosity you can say well what are we curious about well we curious about causal things because we believe that in this specific world in which we causal things will generalize to future props we will talk about which things are relevant for us but i'm not interested in information generally i'm just interested in the kind of information with which i can give a great talk so so so basically for in and and curiosity within machine learning which is a huge field with like hundreds of thousands people of people working on is all built based on different people making different assumptions at that interface now here comes kind of like my difficulty at some level with the keynote now like at some level we've seen an ancestor talk about anzac solo achieve now the answers isn't different to like how anyone in machine learning uses it this we can write the same problem in a few different ways for example it's been known since the 60s that we can convert optimal control problems into equivalent inference problems and vice versa the interesting thing so so that's just like the same answers just written slightly differently so so so from my perspective it's not the answers

per se that is important because where utility maximizes expected value for the future we all agree on the answers what we don't agree are the pieces what can we put in what are efficient algorithms and so forth so at some level the thing that is missing for me in the approach is pushing these things at the interface what are the assumptions about the world what are the assumptions about their cost functions what is what exactly can i say about computation and bits and so forth and that then in a way like no in that talk dr frestren kind of relates to kind of any everything that is close to my heart now let's let's let's let's be positive about that now like curiosity sentience planning self-evidency shared narratives it has like all these things that mean so much to me but we've only ever seen the answers for all of them we haven't seen okay how much does it buy us to make this assumption versus this assumption so everything that computer scientists are concerned with is in a way hidden because we only ever get to see the answers and i'm sure that that there will be at least a minute maybe more than that of rebuttal to what i just said thank you for letting me comment on it back to you gal in two minutes you can take three that's really generous okay all right okay i'll split it one and two so i think i think there are two key points here um the first one is you know the status of naughty theories i think there's a very general and important point there um for me a normative theory would be you know being able to write down an objective function from an engineer's point of view you know if you can write down the objective function then the game starts and then you try and find the best way to satisfy that under constraints and the the perspective that the the free energy principle um that sort of backstories the active inference implementations it takes is um slightly different in fact fundamentally different it takes the point of view of a physicist and say if this system exists what is the objective function so if i was a you know um a classical physicist i'd be looking for the lagrangian and what's the way of um articulating the behavior systems that um um optimize say their action or their package global lagrange and well those are just principles of least action so it is no mistake that the free energy principle is sold as a principle of least action because what is it saying is that given this system exists then it must possess this uh lagrangian that has an interpretation given a markov blanket as um as a sort of variational free energy that you'd find in uh you'd find in um say machine learning or statistics generally so i think that's a really interesting philosophical issue i will leave that and not say anything else the last minute will be the second i think i think pragmatically more important point that that conway was making he was talking about getting the details right getting the the getting the questions right in terms of an actual application of something like the free energy principle to a particular problem i think it's absolutely crucial and for me that

really is a distinction between a generic um application of the free energy principle to what well it's the generative model that's where all the difficult questions lie the structure of that generative model the complexity ways in which you prune it is it hierarchical what kind of message passing scheme is going to be in play out for this kind of generative model continuous visit i've you know so i'm i'm really acquiescing to to conrad's um sort of observation and um that there is something quintessentially vacuous about the free energy principle because it is just a method it does not give you the answer in the sense that this generative model is fit for purpose that satisfies exactly mark's complexity constraints yeah i was just thinking you know conrad do you talk about some simplifying make it more efficient uh which is exactly you know the complexity part that you would use in say basic model selection but the point i'm making here the free energy principle is just a method it's not a solution the solution is actually identifying the generative model that is appropriate for this situation you know in this area of study thank you carl uh sean i think you are the one to introduce yeah it's uh my pleasure to introduce uh dr lauren ross who is an associate professor in logic and philosophy of science at uc irvine so we look forward to your commentary uh lauren great thank you very much um this was a um great talk and it's a pleasure to be here i have two questions and the first one has to do with causation the second one has to do with constraints first question is about the type of causality that's most relevant to this work in particular the type of causality that's present in the generative model itself um and perhaps also the type that they suggest that humans have in mind when their reasoning or an organism has in mind when they're reasoning about the world um so i'm a philosopher of science and of course there are heated debates about how to understand causality and the account of causation is relevant to in particular these generative models there are at least two accounts of causation that are often discussed in the context of generative models one of them are process accounts and the other are dependence accounts a process account of causation is viewing causation in terms of continuity typically that there's some kind of continuity from cause to effect this is sometimes specified in terms of a spatio-temporally connected process and in other cases the transmission of something from cause to effect sometimes it's referred to as a mark marked transmission entity that's being transmitted can be material or it can be energy there's kind of flexibility with what is being transmitted the emphasis in this account of causation is that there's the transmission of something and often that there's a kind of physical connection through the causal process or from the cause to the effect now the second type of causation is a dependence theory in this framework causation is understood in terms of dependency relationships so effects are um outcomes or factors that

depend on their causes and in this sense changes in an effect depend on changes that you would make do make or could make to the cause and you can be completely ignorant of or indifferent to the transmission of anything from cause to effect you don't need that physical connection or you don't need a transmission of any kind of material or perhaps anything else besides causal influence the dependency accounts are also related to control in a number of cases the sense in which we think causes control their effects or that causes are a kind of switch that you could manipulate or intervene on um and that when you do that you get control over the over the effect so the question here is do these generative models have implications or suggestions for the account of causation we as humans or that an organism is using when they're reasoning about the world um and additionally and distinctly what type of causality is involved in the model itself and then the second question about constraints i'm very interested in and i very much like the role of constraints in this framework there is um right the mention of constraints comes up in a number of places they are it sounds like often cited as things that are limiting a kind of space of plausible brain architectures that's how they sometimes come up and they can include things like real neuroanatomy maybe anatomical connectivity um maybe these are empirical constraints but then the generative model is also cited as a constraint um so philosophers of science are very interested in they've become very interested in constraints they interested in clarifying what they are in particular what role they play and explanation and understanding and part of their interest is motivated by the fact that they seem like a unique explanatory factor they are not always causal it seems like maybe in some cases mathematical or they involve a kind of law of nature they're thought to explain limitations in a possibility space which is different from how we standardly think of causes or kind of run-of-the-mill explanatory factor so the second question is about constraints in your um framework and in this model if you consider them to be causal non-causal or if they have some other character and just for you to say a bit more about the role that they play in the model and perhaps also an they're back over you carl for however much time you want that's dangerous i could stick to the three to five minutes no more go ahead girl it's a brilliant question and also artfully um described to the majority of us who probably don't know what a philosophically uh cause and constraint means but that was very limiting thank you and so while it's fresh in my mind the the notion of constraint in this instance um i would read that in uh you know with my physicist hat on um as exactly the way you described which is a um a way of describing disallowed or highly unlikely states of being in some spa possible space of uh states of being that just is um the generative model so um i've been talking about sort of

active influence and self evidencing with it with a strong technological license and uh um cheek basically um you know from the physics point of view that teleology is just not there you're all you're trying to describe is the uh the conservation laws or the implications of certain conservation laws that lead to uh principles of usually symmetry principles like principles of least action and what that basically means for describing something first of all you have to say you know demarcate the thing from everything else and that induces markov blanket but once you've done that then the shape of that thing and the nature of that thing is just basically described by um the sets that uh or the kinds of states that that thing can occupy in the things that it can't so the whole thing is predicated stipulatively on a constraint specification of thingness and then that from that you then interpret put a teleological interpretation on that constraint as a generative model and if you've got a genetic model you've got model evidence and then you can tell a story about self-evidencing but in reality it's you know it's just a you know a mathematical as a calculus just to write down the shape of things so the shape is the constraint and you can describe that probabilistically in terms of a generative model it's literally the probability of causes and effects joint distribution over causes and effects where you're very careful to specify what you mean by cause and effect and that's why you need the markov blanket because you know what's causing things in me well it has to be stuff that's not me so then you have to have a separation so we you know i'm just saying uh introducing that um distinction between cause and effect as something which is quite a big move that you've made as a physicist um because you have to actually write down how would you discriminate between a causal effect and you know then you get into the game of the markov blanket and the second thing was really interesting i i would love to hear more from my perspective and it is a sort of colloquial and possibly limited perspective it sounds as if the two different kinds of causality are essentially those that are of the kind that a control theorist or dynamicist or dynamical systems theory or a physicist would talk about where um it's not so much important about space it's so the temporal continuity aspect that is the if you like the defining feature of dynamics um holes causality so as soon as you write down a launch van equation or a state space model or a markov decision process or a hidden markov model anything that involves time any mathematical process that involves time then you've got causality baked in in exactly the way that you describe it either in terms of marked transitions or transition probability matrices or sort of um you know flows um implicit in state space models or as you know differential equations all of them at the end of the day are core causal structures um in the um in the control theoretic sense simply because you have

described the system of interest in terms of either discrete or continuous time equations so you're talking about dynamics so there is no other um you know causality um your your generative model if it's a generative model of a process that unfolds in time it is by definition a causal model inversion of that model basically it makes inferences you know of a causal sort selecting the right model to account for that causal architecture we'll talk about in terms of evolution or conrad finding you know getting the getting the um doing it under the right kinds of constraints that is just basically um using this principle of least free energy um as a way a method of getting at that causal architecture by inverting a causal model which begs the question what's the other kind of dependency causality i can certainly see a sort of a glimpse of a connection with sort of um pearl like causative static models but you have to remember that that kind of do calculus um and that kind of model that you'd find in things like structural equation modeling and your his interventional calculus um unashamedly about static systems and unashamedly about usually directed a psychic grass that don't exist in the in the you know in the real world um i'm not quite sure about that dependency status um and of course if you do use that dependency sort of causality which we all do when we report a correlation coefficient as scientists i think you're a very shaky ground uh from the point of view of the physicist who just deals with the processes that are not background free because they have time in them you know if you give a physicist time uh then you you've also given him causality so that would be my take on that uh yeah and but i'm not sure about those answers it'd be great to discuss that uh thank you carl before introducing randy just a quick comment uh on the uh on the the the first uh three commentary i was just finishing uh reading a book last weekend it's an historical novel uh the touring alan turig and he is he was a theoretical mathematician physician then moved to do whatever uh he did in his life in his young life and uh one of his regret that the engineer had taken over the basic mathematician in moving forward in understanding decision making so hearing uh mark saying that everybody not everybody but solving the turing puzzle by computer is uh is doable hearing you cal saying that uh you as soon as you have the production uh a production function you can go ahead and optimize what i'm asking to everyone is what if you were being the same at the same time your heart of philosopher mathematician and physician and understanding trying to understand this real-world behavior in real-world context so that's kind of my commentary as we get to the second round of this in mind and also that commentary allowed me to introduce a very uh directly my canadian colleague andy mcintosh who is the director of the institute of neuroscience and the whole technology and the professor at simon frieza uh university and

very much as a normal computational neuroscientist very much moving in that direction of trying to see how we can bring this precision convergence randy go ahead thank you lorette uh hi carl um this is almost like a phd dissertation defense i'm thinking so we have to decide at the end whether you pass or fail or there's a heavy revision i'm appreciating the discussion because i'm getting different perspectives on on the whole collective of optimization and objective functions and generative models and how we can think about using them to address different phenomena and um what i want to try and maybe throw out there for for for discussion is and this you just mentioned this carl about time being a factor in building these generative models so let's take into consideration for instance different time scales over which which events that affect us can happen so there's this event now which is happening over this the time scale of seconds and then our interactions which you you illustrate at the end of your talk um there are uh slower time scales that evolve there are sort of circadian rhythms we have very you know um salient events for example we're all sort of sitting in the middle of this pandemic which has a scale over several years probably going to have a cascade effect that that happens over subsequent years so there are social cultural factors that obviously impact our own interactions impact our physical wellness but also our social dependencies and um you know this gets into the notion of multi-scale models and how to envision these dependencies in some sort of generative model which could i guess use free energy as one of the methods but i guess the question is whether you can build in that time scale interaction where free energy has to span both within temporal scales but also between temporal scales and whether we start getting into this issue kind of like it's turtles all the way down then because we're kind of looking at using exactly the same for different types of generative models but the general model we're talking about one case is more biological whereas another case might be more sort of psychosocial and i'm trying to sort of reconcile um how to think about these various time scales in the framework that we're sort of evolving over the course of our discussion today so i'll stop there and we'll uh let the candidate reply don't say this randy sorry he can't say that because i don't actually i haven't actually got my phd yet ah seriously i didn't quite go around to it so so i get one for precision conversion so that would be quite nice yeah you have time coral yeah yeah now that excellent question i i can see you know that that's that question is also going to come up in the context of sort of natural selection um you know later on um so i think a really key point um and i'm sure you know the answer that the the importance um of the a commitment to a first principle account um is that very much like most of these principles that the they're fairly scale free or scale invariant which does not mean

to say the same free energy $\mathcal{F}$ or whatever other function you want to $\mathcal{F}$ is optimized at all scales what we're saying is the functional form the method you know what it looks like in terms of you know what is a function of what and the shape of those functions that strictly speaking lagrangian is actually conserved over different scales so a lovely example of that in statistics is the fact that if you're trying to optimize let's take a $\mathcal{F}$ let's take you know let's take a variational auto encoder for machine learning that you know is built to change its weights to optimize the elbow the the variational free energy so it's a canonical state-of-the-art $\mathcal{F}$ 21st century free energy $\mathcal{F}$ optimizing machine $\mathcal{F}$ but now say i uh come at this $\mathcal{F}$ i've got a choice of four or uh 400 very short encoders some of them with six levels some with different kinds of uh rectilinear you know activation functions you know different kinds of parameters and i come at this with a model selection or a selective or sort of you know using say differential evolution or a genetic algorithm so what i would then do is score the $\mathcal{F}$ the quality of each of these variational auto encoders and then i would apply a genetic algorithm and do some sort of your selection i would do simulated evolution natural selection the key thing is i'll be using the same objective function in the sense that i will be adding together the elbow or taking the pathogen over the time that i've the temple scale that i was doing my selection and that would be known as basic model selection in statistics so $\mathcal{F}$ the key point here is not the same quantity that's being optimized in the moment as i'm inferring or i am learning and but by taking the path or you're lifting that to the next scale it necessarily has the same functional form and therefore gives the scale freeness to the next level of optimization which in this instance could you know i i framed as uh bayesian model selection or structure learning medical constructivism uh if i was a biologist this would look like $\mathcal{F}$ natural selection $\mathcal{F}$ where i'd now be reading the free energy as a dump to fitness just simply self-evidencing maximizing the likelihood of getting this kind of phenotype this structure in this eco niche $\mathcal{F}$ you know doing this doing this kind of $\mathcal{F}$ self-evidencing or $\mathcal{F}$ behavior and so then formally speaking what that requires you you really to to write down is a renormalization group and practically speaking i think i think the question has the answer in it that whatever principle you bring to the table has to be composable with the separation of temporal scales $\mathcal{F}$ in the sense that you know i can do i can be an artifact that optimizes extremizes my free energy over the next 200 milliseconds and i could do be doing sort of perception it could be over the next week as i learned to ride a bicycle i could do procedural learning it could be over the next three years and i could be doing neurodevelopment or it could be transgenerational i would be doing $\mathcal{F}$ i'd be doing natural selection but at every point $\mathcal{F}$ each of the the

objective function has has if you like a mapping via the notion of this um scale freeness and so that the optimization at one level is essential for the optimization of the level below and vice versa so if i was like george ellis i'll be talking about top down and bottom up causation now which means i i have to have the right heritage as a phenotype in the right species in order to have the right kind of brain in order to do the right kind of learning in order to make the right kinds of inferences but on the other hand i need to do the right kinds of inferences to learn the right things in order to do the right development so that so i have the right adaptive fitness so i can be selected so there's a circular causality that couples each of these levels now your question is can you usually write that down in silico and start to actually grow you know intelligent artifacts or simulate and model systems that are subject to these mutual constraints in the circular causality that's a really difficult problem and particularly when any one level starts to show things like curiosity and the like you know the computational complexity is is has to be met but it is great this is a challenge and then we come back to complexity minimization keeping things as simple as possible right thanks carl great and before i introduce our next uh panelist i want to remind all the committee members to return your form about the dissertation so that we can turn that into the administration it's my pleasure to introduce dr martin paulus uh who is the scientific director and president of the laureate in institute for brain research in tulsa oklahoma so uh look forward to your commentary martin thanks sean and uh thanks carl um it um and i i've ever really actually quite enjoyed the um i have to comment on all the other comments before i get to my uh i really enjoyed what uh convert was saying and i think that resonates very strongly with me that uh recon was conrad was highlighting that um you know the free energy principle and active inferences and unsats it's a start to think about something the difficult work is actually trying to find what's the right what uh called what you were saying what's the lagrangian for the things that we're interested in how is it being implemented i'm going to say something before i ask the question because i also want to make sure that i understand what you're saying so let me say a couple of sentences and then i want to formulate my question um i understand active inference is a theoretical and computational prescription for normative behavior in complex environment that is fundamentally based on bayesian influence and as stated by you prime probability distributions are used within the constraint of the free energy principle to arrive at posterior distribution that quantified the behavior to be observed that is most consistent with the underlying belief about the nature of the environment so that's my understanding as i would say and so but as a as a so now i'm putting my my psychiatry head up you know um i'm into

uh i'm interested in alteration or dysfunction of behavior if i'm correct with my premise of what i just stated and if you you can correct me and i could be wrong and that's what i want to but if i if that is sort of the correct premise so alterations of dysfunction of behavior can be conceived in this framework in this anzacs um as extending from an altered and possibly inaccurate belief about the state of the or by a faulty integration of available to form adaptive posterior distribution um and so my question um that i wanted to kind of just get your input on how might one arbitrate between those two different um alternatives um and um and and maybe more in the biology how might these different differences be implemented but we probably won't have time to discuss this um because i also actually think and now i'm coming becoming sort of more real practical um um you know because a and b could mean different things for remediation or intervention right because one of the things that we want to do if peop if people because i'm a psychiatrist or animals if if you're an animal person uh who behave in such a way that we would agree in in some form is non-adaptive we want to help them to become more adaptive but if the dysfunction is based on fixed prior probabilities about their environment or faulty computation the the remediation might be actually quite uh different and therefore you know we need to find almost ways of arbitrating between those two alternatives and so i'd like to get sort of your perspective on that and how would you go about doing that thank you thank you i'm sorry um yeah well you know how how i do it i do it like you're doing it let's actually ask a cheeky question um so um the way i resolve this mathematically now it's a really interesting question but i i i sort of deflate the question the problem is you know is this broken inference um or just bad prize yeah bad prize with with with great message passing schemes or broken message schemes with good prize and that's a really interesting distinction but practically i don't think it's terribly um it's it's terribly meaningful from a practical point of view um so i dismiss that with something called the complete class theorem and the complete class theorem just says for any pair of observable behavior and some loss function there are some prior beliefs that render that behavior based optimal so what that allows me to do is say that everything that's broken can be cast as a as a an unusual prior so that this prior so unusual that it looks as if it's a faulty message passing system so i had a prior belief about attention deployment and precision and while i'm here just open brackets of course this is a lovely connection between precision convergence and precision in the sense of mark and deploying attention you know and indeed from uh martin your point of view your deficits of attention and deficits of consciousness are implicit in deficit attention so precision here getting the precision right these sort of um inferences about

the variability or the um the width of probability distributions in terms of the encoding of uncertainty say that i had bad prize about that what would that look like well it would look like i had an attention deficit disorder that would look as if i had faulty evidence accumulation and assimilation that would look as if i basically had a broken imprint scheme so what i do is i i don't worry about it what i say is the real job and then you come back to your second i think key point um the real job is to find out what part of this deeply structured hierarchical generative model that is a genetic model of dynamics um is broken sorry not it is broken is uh and that is described by um by its priors uh and formal prize in terms of its structure what part is abnormal yes broken in in this kind of person or this disease relative to another another disease that's i think the question you're asking of course it is exactly the practical question that conrad was referring to and just to pick up on one um you know your excellent summary of of active imprints um that is if you like an application the free energy principle you're absolutely it's just a prescription it's a prescription as a method of waiving things so just like hamilton's principle of least action it's a method so you give me a particular ball and a particular sort of cannon and i'll tell you the trajectory but before you do that there's nothing for me to do so all the heavy lifting and all the hard work because it's basically finding the generative model now under the assumption that you know you can now apply the free energy principle variational inference of an inactive sort or active inference then that provides a good account or explanation of this person's behavior then the job is just the normal clinical neuroscience problem of finding what part of this charity model is different and by this part i mean the particular prize and usually hyper prized and they're usually um as you will you can't respond because it's my turn to talk but what marty would like to say they're the kind of prize and there are these higher order prizes whose biological substrates are probably tied up with um neuromodulators and synaptic excitability and all the physiology that goes hand-in-hand with that sort of fast synchronous interactions and the like so a very specific class of uh of part of neurobiology that deals with with uh modulators and particular kinds of neuropharmacology and this i think is a perfect example of actually sort of saying what of the universe of this deeply structured genetic model that um you know is not only um describing a person but in a cultured context what's remarkable about this prior relative to this normal typical uh prior and that's a really difficult game obviously yeah i actually had a second so so even though this has already been a very uh fascinating discussion for me i actually had a second question because we were asked to have two questions i do i want to that's a little more edgy not edgy for you but edgy maybe in in the sign of times um which is the fact is and you

you actually have now expanded which is very uh i i i just can't wait to see what you're going to do with the message message passing scheme between active inference agents right because now we're talking more than one agent and you you gave the the nice examples of the 20 questions and how one can develop a common framework to then arrive at the correct solution but of course again coming back to the fact that i'm interested in when behavior goes wrong we live in a time first of all let me just say that you know even psychiatric disorders or abnormal behavior individuals are not islands they're embedded in in communities and these communities interact heavily and therefore doing the kind of message passing that you're talking about but coming back to first principles you know we live also in a time where it seems like that some agents together are very good in message passing and other agents are very good in message passing and arriving at some consensus but then there is nucleations that seem to be occurring and they seem to be getting more prominent and more severe in in the current time where there is very little message going between those nucleations and so my question would be um how how would you understand that as uh emerging uh from uh sort of an active inference principled agent that interacts with other active inference um principled agents yeah that's a great question and i suspect it probably will probably just use that question when we come back to sort of round and summarize one of the ways forward the way that i would do that is basically by numerical analysis and i would write down a little world and see what emerges when each member of a population or ensemble you know is described or its behavior is prescribed by uh by active inference and what i would imagine would happen is this dialectic between trying to make your world as predictable predictable and at least surprising as possible um whilst at the same time trying to share your genetic model with as many artifacts like you that are generating the things you're trying to predict will be resolved by this nucleation and this sort of blanket building um isolationism if you like a sort of partial isolationism um that characterizes so many aspects of our sort of political and geo and familial cultures and structures that there is again a scale invariance that does entail um sort of a markov blanketing or blanketing or segregation or partial segregation that fulfills that within your blanket within your family within ethnic or political or religious um you culture there is a shared narrative and everything is nice and predictable um and that that that sort of encapsulation is underwritten by something we're referring to before which is this bayes optimality in terms of bi-sampling you know if i act to realize the kind of world i think i'm built to uh self-evidence in then i'm going to construct into the extent that i'm going to be successful in so doing collectively then those boundaries will um will will be maintained the

question now is of course you know what is the scale the structure the number of these of these communities that are emerging properties very much in the spirit of turing pattern formation that you know we're talking about uh i was talking before about alan turing i mean this is exactly the kind of segregation reaction diffusion-like phenomena um he wanted to understand and i would imagine that the same techniques will be used uh a century later and by which i mean he started off by basically doing by hand excel spreadsheets and looking at sort of local exchanges to look at your self-organization and pattern formation we would do it in computers but actually probably just about the same speed because we'll be actually modeling active inference agents that actually are modeling everybody else in their eco niche and who knows what will happen but hopefully for a sufficient number of scale invariant aspects to this procedure you'll see what tends to happen in your own brain that you actually get functional specialization in the context of integration um so you get distributed processing but there is segregation of function um and you know you can chase that right down to the structure of a neuron or right up to the structure of your body there are symmetries but there's also specialization so i imagine that there will be a similar kind of structures emerging that will be describable in terms of the dependency kind of causality through markov blankets that one could be looking for in silico thanks uh carl in the interest of time you're right what you just said is essentially the overall round table but in the interest of time i would like that we uh intro we get the commentary of the la of the last two presenters and you integrate your reply into we will start our round table with you cal for the last so if it's okay ally and uh daniel will be uh presenting their commentary and then we'll start the the round table uh with you and your answer to those two at the same time of the general question how do we go for this real procedure called results sound back to you to introduce alain it's my pleasure to introduce dr elaine dagger who is associate professor at the montreal neurologic institute and mcgill university a neurologist specializing in movement disorders and functional brain imaging and little known a fantasy american football aficionado you so thank you uh carl for a really inspiring talk and uh i i have two related questions or comments one has to do with what you refer as your day job uh functional neuroimaging we've been able to measure blood flow in the brain for a few decades now and in the early days we just asked questions like what does what but i think the more interesting prospect is to try to measure information in the brain and specifically to ask how is information processed in the brain and one way that you have promoted is to try to quantify cognitive processes and then relate them to brain activity and i wonder what you think the prospects are for for this area and specifically i'd like to ask you

about dopamine which encodes information in multiple ways and i'd like to specifically ask you about tonic dopamine because it seems to be related to motivation into the willingness to act in the world patients with low dopamine low tonic dopamine so that's the dopamine that basically bathes neurons these patients experience apathy despite the fact that they still enjoy life and they still have the ability to experience pleasure they're not willing to act and conversely when we increase their tonic dopamine with medications we can trigger states of maladaptive action such as pathological gambling and compulsive sexuality and i was trying to think where tonic dopamine fits in the free energy principle and i was wondering whether it could encode expected information and that that alone could explain uh its its effects on apathy and motivation quite interesting uh do you have another question elaine before i introduce that no that that's the that's okay very good uh in let's shift to uh dandelski who was wondering why that we had invited him on the panel and it was very very carefully uh selected for donnie's working from the genes to society is working on issue of aging life-course socio-economic status that uh are so important if we are to create this next generation science of real-world behavior in real world context in a more adaptive manner so dan is assistant professor of epidemiology at columbia university uh melbourne school of public health and uh also um at the robert butler uh columbia aging center so down what i don't i let you to all the space we can get go ahead well thank you so much um for having me for the invitation uh it was a fascinating talk uh and and equally fascinating questions from the panelists uh i've learned a great deal um and in fact the questions that i initially wrote down to pose to carl uh which were about how this model speaks to the development of individual differences and in particular uh how that happens across uh time scales of human development um across the lifespan have been partly answered in in the responses in particular to randy regarding time um and to martin uh regarding uh the genesis of individual differences so in service of transitioning us toward that overall panel let me pose the more general question to carl from the public health perspective since i'm here from the department of epidemiology and i know you've um uh been thinking uh about public health and epidemiology in the context of your models uh recently um how should the free energy principle inform the kinds of strategies we are developing to shape public behavior private behavior in service of of cultivating health for individuals across their lives and that's a that's a um you know maybe the the opposite end of the spectrum from alan's question which is you know very specifically about a particular mechanism that you've thought about and i guess i'm asking more generally not what does this say to you know sort of public health in general but what are applications that you might

envision of the free energy principle that could guide us toward a more precise approach to public health and social policy um than where we stand today um and perhaps you know we can have some further interaction in the subsequent q a thank you very much don and i would like to make sure that all the panelists are having their video on we have uh 12 more minutes in those 12 more minutes we will have a cal addressing those first two those first two uh uh last two commentaries but within the broader perspective call of your word of wisdom how can we as a worldwide community move toward this next generation precision convergence uh convergence science so carl has two or three minutes and each of you will have one minute after and then sean and me will be uh concluding go ahead calm thank you very much your question and elaine's question are actually i think of of the same milk and like really important and also speak to uh what we were discussing with martin which is you know you've got um a principle a recipe how are you going to make it work in a useful way in different scales um so you know the in answer to to daniel's question um one can um drill down on the the the individual on the single person in terms of their neurology and their psychology um as an individual and try to understand the functional architectures and the physiology of the belief updating and the um message passing neural message passing very much along the lines that we're discussing with mark and with martin so this basically would be uh understanding neurology psychiatry and indeed behavior as aberrant belief updating from the point of view of active influence but because you've got this process theory underneath the hood that that grounds into neurobiology you've also now got a way of articulating psychopathology in terms of pathophysiology things that are you know that are aberrant and highly unlikely in terms of the physiology and the dynamics and the structure the neuroanatomy of a particular brain so that you know you that is if you like you're already an active research part of say computational psychiatry and computational uh neuropsychology you just build things in silico you see you make sure you optimize them so they behave exactly like your patient you do a bit of precision medicine just by nuancing and adjusting and optimizing the price of this digital twin if you like um that complies with the the method of the free energy principle then i think you have to move up to the next level which we already talked about which is the um the exchange between agents and issues data foraging not data so i meant sort of information foraging epistemic trust static interactions therapeutic engagement uh martin if you had time would speak to this you know how can you use these principles to predict the kind of prior beliefs that people will have will commit to um that ensures a therapeutic engagement with the doctor say or with the public health service these are really important issues um to understand

the mechanics of the belief updating in terms of interpersonal inference may be really important in just supplying public health care uh let alone you know what that care the content of that care and then at the next level we're talking about epidemiological modelling um but we haven't done that yet in the sense that when we model viruses and model people the vectors of those viruses the people are just like points like this the physics spherical cow that they don't insert in themselves have any um uh intentions or prior beliefs that's an outstanding issue and i repeat i think that's where we're going to end up i won't finish that because i want to just comment alan's that's a great idea i haven't thought about that sorry dopamine may well be the encoding of the expected information gain of the you know the epistemic value of that's really very interesting i think it's entirely licensed because um speaking to the exchange with martin um what we're talking about here is prior beliefs about the precision or the confidence or the certainty with which you sub personally your brain holds certain beliefs that means for every probability distribution every probabilistic representation in the brain there is a precision that may or may not be encoded implicitly or explicitly with a neurotransmitter so that certainly speaks to a plurality of things that things like dopamine might encode we certainly know something to do with plans and action but beyond that i think that the hypothesis it could be expected information gain is exactly the same kind of hypothesis that you would then use simulations to test by you know playing off expected cost for example against expected information gain factorially and seeing which was the best account for dopamine in this region as opposed to dopamine responses in that region that's a excellent idea but would be part of this basically building a brain in silico optimizing it using the principles we've been talking about to you know to drill down uh you know in a precision medicine kind of fashion um which would be one end of the scale from a public health perspective the other end of the scale being where i'm going to finish which is the agenda that days do you uh you want to go with your one minute commentary on the where we go from here well in a sense i i the last the last 30 seconds was was meant to be you know i think we've identified you know what where the challenges are and it's basically something which everybody's spoken to some level from conrad how you have the complexity of it all from randy you know how do you get the multiple time skills from you know daniel how do you how do you get this to the level of public health it's all about basically self-organization of sentient artifacts at different scales so it is exactly collective agency so that's the big challenge uh the last thing i'd like to say is i i hope i get the uh the phd that'll be quite clear oh you do absolutely you get many post doc so let's let's have this 30 second to one minute commentary for each of you you can you can on

vote other phd but most importantly i want to hear from you where do we go as a capacity building worldwide community in the next six months of the year uh mark yes i will follow the same order than we had in the panel mark so alan turing keeps coming up in the conversation today and i don't think that's an accident turing's world view included the theory of computation and i don't mean digital computers in your laptop i mean the the theory of information dynamics as a unifying theory of everything and it permeated all of his scholarship and i i you know i don't want to talk my own book here but but i think the theory of computation offers fertile basic ground just like set theory did a couple of centuries ago for mathematics to create a common language and a common set of models i think building this the underlying theory for machine learning is very very important and more broadly for neuroscience which means if we're in these normative spaces oftentimes our algorithms do a lot of arbitrary things and i think there's a lot of wonderful recent developments in the area where you can then prove that say you will generalize for a certain set of phenomena like pac-bay i'm thinking of ideas like pack bass and and and i i think not like and they interface with theories of how the world was now they they depend on all these pieces that we come in that we put in and i think there will there's there's a lot of development that we need to make these ideas be concrete thank you lauren i mean what i take away from this is that um there is a lot of work to be done and thinking about how to build the generative models that we need to answer some of these questions i think as conrad posted some of the examples of the application of free energy in the domain of cobit modeling we've done something similar um using merging agent-based models with some sort of sir models and that's also a generative model just a question of the objective functions i think this is a domain that kind of extends beyond just thinking about the the tool itself but thinking about the application and where the application can take us and that's i think been a constant theme um so it's not just so much thinking only about free energy per se but also you know what do we want what do we want to understand and can we use the free energy to to engender new understanding of these problems thank you martin kevin very brief i want to thank carl for challenging all of us to think harder about you know how behavior emerges and how it breaks down again i'm excited to see what multi-agent active inference uh will look like and what you know what patterns emerge and how we can potentially use the insights to make the la the world a better place thank you alain okay thank you well i think that the challenge is to see uh what is a vulnerable brain at least in medicine and and understand how genes and environment conspire to create a vulnerable brain and to understand that vulnerable at a at a computational level so that's the

interesting challenge of the next few years done i mean i think i would just echo alan i think i think i was inspired by mark's initial comments about the idea of information as a fundamental element in the universe and genetics as an encoding of that information and so the way in which um that information interacts with the inner information of our experience to produce variation and behavior is is the is the frontier um and and i i should also just say i'm grateful that lauren has returned and so uh oh boy um well yeah so i i mean carl's work is already bringing together many approaches and i think this is particularly useful for many questions especially ones that are kind of on the edge of discovery and in some sense the edge of science so where do we go from here i think we can um kind of capitalize on this and do more of this having more discussions asking more questions and more attempts to answer them so thank you shifting to you uh in fact lorenzon uh shifting to use on almost use the expression you are using what we are doing is brings bridging the edge of science i will let you speak for yourself or your commentary but also introduce our next precision convergence uh that is in fact a very nice buildup to what we were having uh today go ahead sean yeah and i gotta do it quick because i have to go meet my dean uh but thank you for the wonderful talk carl and thank you all the panelists for a mind-expanding conversation i feel like uh you know you're talking lagrangians message passing free energy i feel like i'm in a quantum chemistry talk again which is my background but uh it's it's it was really uh interesting and stimulating talk in our next presentation will be i believe on may 24th is putting the poster in the chat thing from mit we'll be talking about uh both basically how neural uh neuroinformatics is facilitated by infrastructure development he is the lead on an infrastructure project called dandy which is a microservices architecture for bringing together neuroscience pipelines and data so a very different talk but still interesting nonetheless but thank you again carl for what a very wonderful yes indeed thanks to carl uh we very very much appreciate uh you placing this in your busy schedule and each of you on the panel and those in the audience also uh we all know what two hours mean and we appreciate very much uh not only the time but the brain investment in the last two hours uh is indeed building this capacity for precision convergence which is this edge of science connecting all of the above real-world transformation in as real-time as possible thank you everyone we will end here uh and we hope to have you on the next one but most importantly um it is it is uh a community we are building so we will be getting in touch in that in diverse manner thanks to all of you bye-bye you